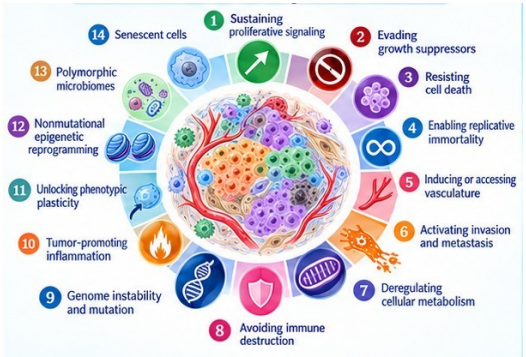
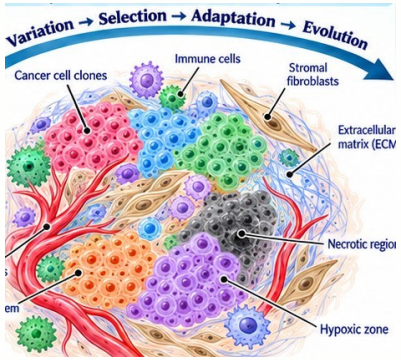
INTRODUCTION · CANCER COMPLEXITY

A complex, dynamic and heterogeneous disease



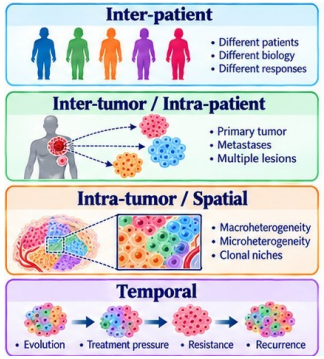
1 Complex biology

- 14 hallmarks and enabling characteristics
- Genetic and epigenetic alterations
- Shaped by the microenvironment



2 Dynamic ecosystem

- Cancer, immune, stromal cells and vessels
- Variation → selection → adaptation → evolution
- Hypoxia, immunity and treatment drive change



3 Heterogeneous

- Between patients
- Between lesions in one patient
- Within a tumour and over time

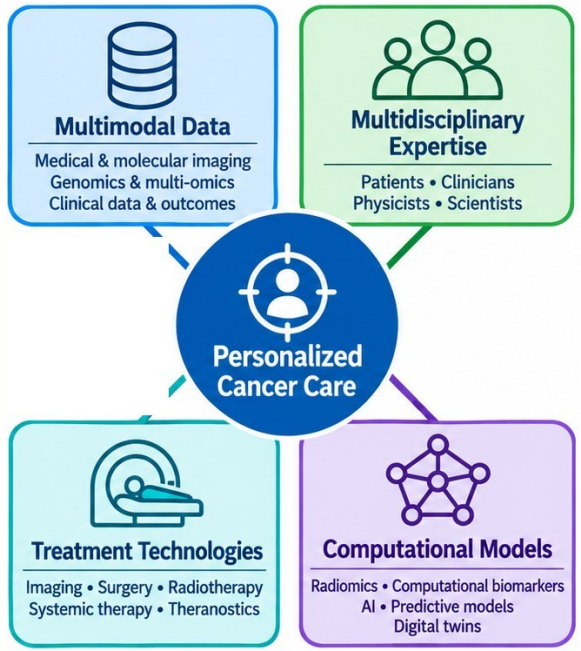
Clinical consequences: variable prognosis · differential treatment response · resistance · metastasis · recurrence

From the care continuum to personalized care



1 Care continuum

- Prevention and screening
- Diagnosis and stratification
- Treatment planning and delivery
- Response assessment
- Survivorship and monitoring

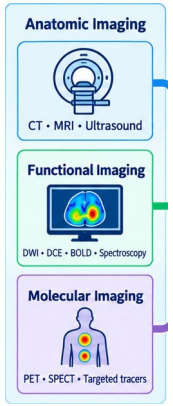


2 Personalized care

- Multimodal data: imaging, omics, outcomes
- Multidisciplinary expertise
- Treatment technologies
- Computational models: radiomics, AI, digital twins

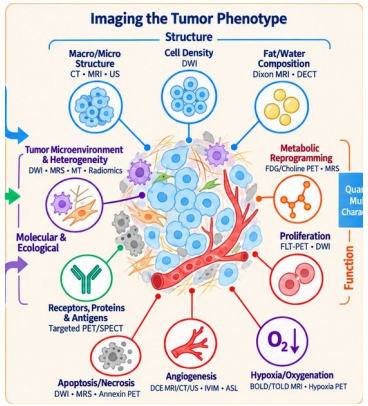
Multimodal · multiscale · multipurpose · multidisciplinary — evidence → decisions → outcomes → learning

Imaging the tumour phenotype



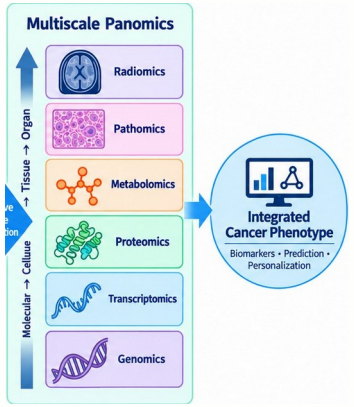
1 Imaging modalities

- Anatomic: CT, MRI, ultrasound
- Functional: DWI, DCE, BOLD, MRS
- Molecular: PET, SPECT, tracers



2 Tumour hallmarks we can image

- Structure: cell density, fat/water
- Function: metabolism, proliferation, hypoxia, angiogenesis
- Receptors, apoptosis, microenvironment



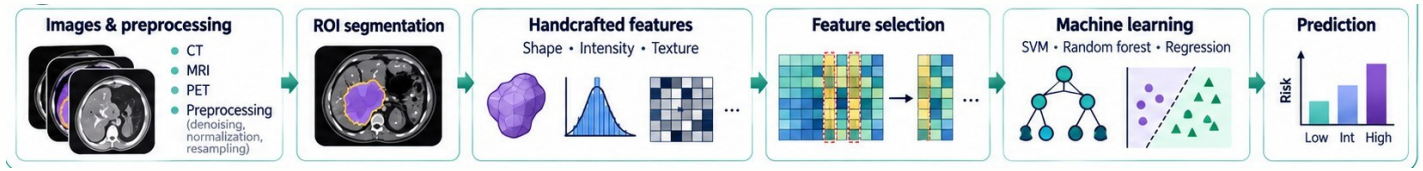
3 Multiscale panomics

- Radiomics, pathomics, metabolomics, proteomics, genomics
- → integrated cancer phenotype

Quantitative, multiscale imaging → biomarkers, prediction and personalization

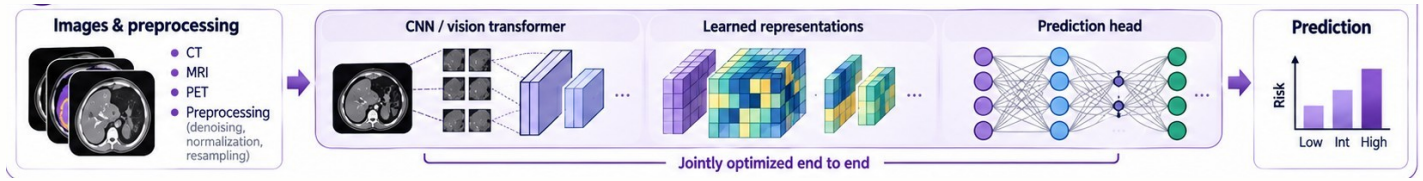
IMAGING BIOMARKER · RADIOMICS

Three radiomics workflows



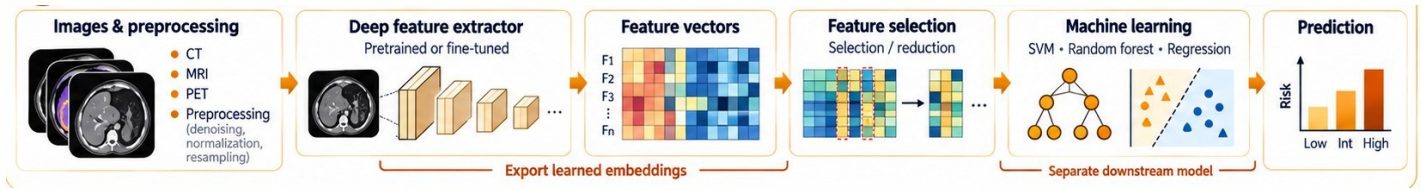
A Handcrafted radiomics

- ROI → shape, intensity, texture features
- Feature selection + machine learning



B End-to-end deep learning

- CNN / transformer learns the features
- Optimized end to end, image → risk



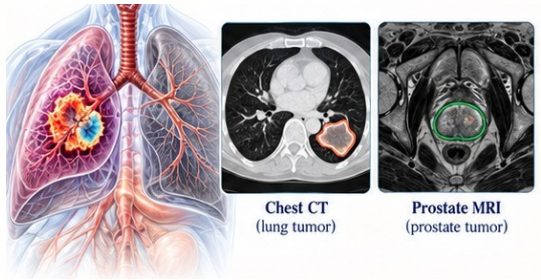
C Deep features + ML

- Pretrained network exports embeddings
- Separate ML model predicts risk

Applications: diagnosis · prognosis · treatment response — independent validation required

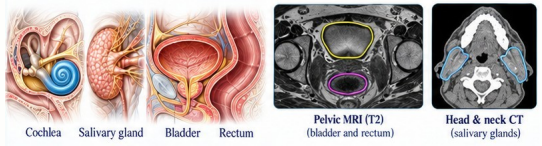
IMAGING BIOMARKER · RADIOMICS

Radiomics: from tumour to whole patient



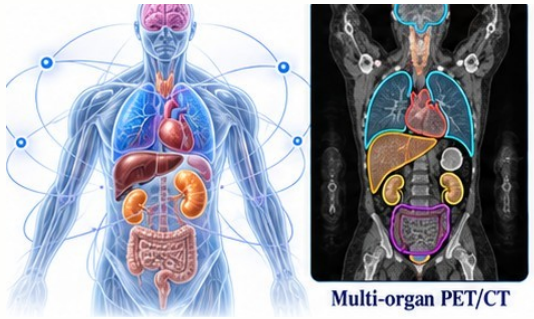
I Tumour phenotype

- Prostate, lung, brain, kidney
- Grade, molecular status, prognosis
- Treatment-response prediction



II Normal tissue

- Cochlea, salivary glands, rectum, bladder
- Changes after radiotherapy
- Toxicity-risk prediction



III Systemic, multi-organ

- Host beyond the tumour
- Endocrine-organ signatures
- PSMA-negative prostate cancer



IV Non-oncologic

- Myocardial infarction (cardiac MRI)
- COVID-19 lung disease (CT)
- Bone mineral loss (radiograph)

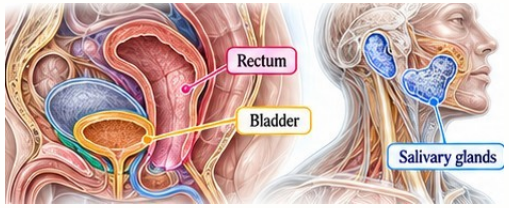
Cross-cutting: multimodal imaging · radiogenomics · longitudinal / delta radiomics

Applications of radiomics



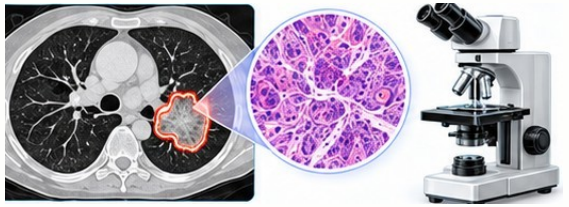
I RT response

- Predict tumour response to RT
- Identify resistance patterns



II Toxicity

- Normal-tissue injury risk
- Salivary glands, rectum, bladder



III Diagnosis

- Tumour type, grade, aggressiveness
- Support lesion characterization



IV Prognosis

- Progression and recurrence risk
- Stratify patients by outcome



V Radiogenomics

- Link features to tumour biology
- Molecular signatures



VI Monitoring

- Track change during and after therapy
- Delta radiomics

Beyond the tumour: systemic and non-cancer disease · needs reproducible features and external validation